# Murray Patterson

## Curriculum Vitae

## Education

2006—2012 Doctor of Philosophy, Department of Computer Science, University of British Columbia, Vancouver, Canada, GPA – 4.0/4
Advisors: Ján Maňuch, Cedric Chauve and Arvind Gupta
Thesis title: Variants of the consecutive-ones property motivated by the reconstruction of ancestral species

2004—2006 Master of Science, School of Computing Science, Simon Fraser University, Burnaby, British Columbia, Canada, GPA – 3.9/4
Advisors: Eugenia Ternovska and Arvind Gupta
Thesis title: Grounding for model expansion in $k$-guarded formulas

1999—2003 Bachelor of Computer Science (Honours), Jodrey School of Computer Science, Acadia University, Wolfville, Nova Scotia, Canada, GPA – 3.8/4 (top 5%)
Advisors: Raymond J. Spiteri and Jim Diamond
Thesis title: Implementing Runge-Kutta solvers in Java

## Work Experience

2020— Assistant Professor, Department of Computer Science, Georgia State University, Atlanta, Georgia, USA
Teaching both undergraduate and graduate courses in computer science, carrying out a research program in bioinformatics, and serving the department, and the university

2019—2020 Visiting Assistant Professor, Department of Computer Science and Engineering, Fairfield University, Fairfield, Connecticut, USA
Teaching both undergraduate and graduate courses in computer science, as well as carrying out my research program in bioinformatics

2016—2019 Postdoctoral Researcher, Experimental Algorithmics Lab (AlgoLab), Informatics Department, Università degli Studi di Milano-Bicocca, Milano, Italy
Advisors: Gianluca Della Vedova and Paola Bonizzoni
Performing research on haplotype assembly, cancer phylogenetics and their application

2014—2016 Postdoctoral Researcher, Laboratoire de Biométrie et Biologie Évolutive (LBBE), Université Claude Bernard Lyon 1, Villeurbanne, France
Advisors: Daniel Kahn and Vincent Daubin
Studying the co-evolution of (metabolic) function, and how it can be used to help the construction of phylogenetic trees

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2013—2014 ERCIM Postdoctoral Research Fellow, Life Sciences Group, Centrum Wiskunde Informatica (CWI), Amsterdam, The Netherlands
Advisors: Alexander Schönhuth and Gunnar W. Klau
Improving the state-of-the-art haplotype assembly methods for next-generation genome sequencing data

2012—2013 ERCIM Postdoctoral Research Fellow, INRIA Rhône-Alpes and the Laboratoire de Biométrie et Biologie Évolutive, Université Claude Bernard Lyon 1, Villeurbanne, France
Advisor: Eric Tannier
Performing a study of the co-evolution of genes, and how it can be used to help the construction of phylogenetic trees

## Teaching Experience

Spring 2026 Georgia State University, Atlanta, Georgia, USA
- CSC 4330 / 6330 — a senior year / masters-level programming language concepts course (two sections)

Fall 2025 Georgia State University, Atlanta, Georgia, USA
- CSC 4510 / 6510 — a senior year / masters-level course on automata theory

Fall 2024 Georgia State University, Atlanta, Georgia, USA
- CSC 4520 / 6520 — a senior year / masters-level course on algorithm design / analysis, and
- CSC 8540 — a grad-level course on advanced algorithms in bioinformatics

Fall 2024 Georgia State University, Atlanta, Georgia, USA
- CSC 4330 / 6330 — a senior year / masters-level programming language concepts course

Fall 2024 Georgia State University, Atlanta, Georgia, USA
- CSC 4330 / 6330 — a senior year / masters-level programming language concepts course

Spring 2023 Georgia State University, Atlanta, Georgia, USA
- CSC / DSCI 4810 / 6810 — a senior year / master-level course on artificial intelligence

Fall 2022 Georgia State University, Atlanta, Georgia, USA
- CSC 4330 / 6330 — a senior year / masters-level programming language concepts course

Spring 2022 Georgia State University, Atlanta, Georgia, USA
- CSC 4330 / 6330 — a senior year / masters-level programming language concepts course

Fall 2021 Georgia State University, Atlanta, Georgia, USA
- CSC 4710 / 6710 — a senior year / masters-level course on database systems

Spring 2021 Georgia State University, Atlanta, Georgia, USA
- CSC 4330 / 6330 — a senior year / masters-level programming language concepts course

Fall 2020 Georgia State University, Atlanta, Georgia, USA
- CSC 4330 / 6330 — a senior year / masters-level programming language concepts course

Spring 2020 Fairfield University, Fairfield, Connecticut, USA
- CS 131 — an introductory programming course in Java, and
- CS 354 — a junior year theory of programming languages course using Haskell and Prolog

Fall 2019 Fairfield University, Fairfield, Connecticut, USA

- CS 131 — an introductory programming course in Java
- CS 231 — an intermediate (sophomore year) programming course, focusing on the object-oriented features of Java, and
- SW 402 — a senior-year / masters-level course on database management systems

2015 Instructor, Department of Mathematics, Université Claude Bernard Lyon 1, Villeurbanne, France

Section head: Raymond Lobry
Teaching a section of mathematics for the life sciences to first-year university students

## Advising

Spring 2026 I supervised PhD students Prakash Chourasia, Avais Jan, Ayotomiwa Ezekiel Adeniyi and Mansoor Ahmed, and Master's student Ramita Rathore

Fall 2025 I supervised PhD students Prakash Chourasia, Avais Jan, Ayotomiwa Ezekiel Adeniyi and Mansoor Ahmed, and Master's student of the Biology Department James Dang

- Mansoor Ahmed won an MBD fellowship, funding the remainder of his study.
- Prakash Chourasia received the 2025 MBD Outstanding Fellow Award.
- James Dang successfully completed his Master's project titled: "Analysis of the chromatophores involved in anemonefish coloration using bioinformatics to identify DUOX1 and other genes in the chromatophores".

Spring 2025 I supervised PhD students Prakash Chourasia, Avais Jan, Ayotomiwa Ezekiel Adeniyi and Mansoor Ahmed

- Prakash Chourasia received the 2025 Outstanding Graduate Student Award in the Department of Computer Science.

Fall 2024 I supervised PhD students Sarwan Ali, Prakash Chourasia, Avais Jan, Ayotomiwa Ezekiel Adeniyi and Mansoor Ahmed, Master's student Sai Roopesh Mandava, and Undergraduate student Ramita Rathore

- Sai Roopesh Mandava successfully completed his Master's project titled: "Kickstarting football skills: A reinforcement learning approach".
- Sarwan Ali successfully defended his PhD dissertation titled: "Designing representation learning methods for molecular sequences analysis".
- Sarwan Ali received a Georgia State dissertation grant award.

Spring 2024 I supervised PhD students Zahra Tayebi, Sarwan Ali, Prakash Chourasia, Taslim Murad, Avais Jan and Ayotomiwa Ezekiel Adeniyi, Master's students Raghuveer Gummadi and Sai Roopesh Mandava, and Undergraduate students Ramita Rathore and Anderson Hernandez

- Taslim Murad successfully defended her PhD dissertation titled: "Designing methods for representation learning of molecular sequences and its application in analysis tasks".

○ Zahra Tayebi successfully defended her PhD dissertation titled: "Clustering, classification, and simulation of evolutionarily related sequences in the context of cancer and viral spread".

○ Sarwan Ali received the 2024 Graduate Research Award in the Department of Computer Science.

○ Ramita Rathore won a 1st Place Novice Research Award at the 2024 GSU Summer Undergraduate Research Symposium.

Fall 2023 I supervised PhD students Zahra Tayebi, Sarwan Ali, Prakash Chourasia, Taslim Murad, Avais Jain and Ayotomiwa Ezekiel Adeniyi, and Master's students Raghuveer Gummadi and Eunice Olorunshola.

○ Prakash Chourasia won an MBD fellowship, funding the remainder of his study.

○ Zahra Tayebi successfully completed her dissertation proposal titled: "Efficient clustering and classification of protein sequences".

○ Sarwan Ali successfully completed his dissertation proposal titled: "Designing representation learning methods for molecular sequences analysis".

○ Eunice Olorunshola successfully defended her Master's thesis titled: "Classifying different cancer types based on transcriptomics data using machine learning algorithms".

Spring 2023 I supervised PhD students Zahra Tayebi, Sarwan Ali, Prakash Chourasia, Taslim Murad and Avais Jain, and Master's students Raghuveer Gummadi, Eunice Olorunshola, Sai Venkata Sumanth Gaddam, Kushal Khatri, Surya Subrahmanya Sai Chandra Sekhar Kolli, Samvidha Reddy Mannem, Fatma Noor, Ashwin Pilgaonkar, Ria C. Thazhe Punathil, Sainikhila Vemulapalli and Karishma Mahendra Wadhe.

○ Taslim Murad successfully completed her dissertation proposal titled: "Designing methods for representation learning of biological sequences and its application in analysis tasks".

The following Master's projects were graduated:

○ Sai Gaddam: "Diabetic retinopathy detection using deep learning".

○ Kushal Khatri: "A stratified subsampler for hierarchically structured data".

○ Surya Kolli: "Recommendations using graph databases".

○ Fatma Noor: "Can community detection build a good phylogeny?".

○ Ashwin Pilgaonkar: "Robinson Country Intelligence Index (RCII) – A country risk analysis tool".

○ Ria Punathil: "CoVScan – A tool for viral host specification from coronavirus spike sequences".

○ Sainikhila Vemulapalli: "Vehicle image classification and damage analysis using deep learning model".

○ Karishma Wadhe: "Drive safe – A driver drowsiness alert system".

Fall 2022 I supervised PhD students Zahra Tayebi, Sarwan Ali, Prakash Chourasia and Taslim Murad, and Master's students Supriya Arun, Kaustubh Khandai, Kushal Khatri, Prashanth Annam, Babitha Reddy Devireddy, Indu Munnangi, Venkata Krishna Chaitanya Nuthalapati, Dhruval Patel, Shraddha Sharma and Atharv Utkarsh Yeolekar.

- Zahra Tayebi won a B&B fellowship, funding the remainder of her study.

The following Master's projects were graduated:

- Supriya Arun: "Comparative analysis of Covid classifiers".
- Kaustubh Khandai: "Determining significant correlation between pairs of extant characters in a small parsimony framework".
- Prashanth Annam: "Neobook: A social medial application using Flutter and Firebase".
- Babitha Reddy Devireddy: "Chaos engineering using Gremlin and GC analyzer".
- Indu Munnangi: "My recipe book: A MERN stack web application".
- Venkata Krishna Chaitanya Nuthalapati: "Tsuba: Web security scanner".
- Dhruval Patel: "Best actor detection".
- Shraddha Sharma: "Sunsetting customer identification service".
- Atharv Utkarsh Yeolekar: "Data condensation using generative adversarial networks".

Spring 2022 I supervised PhD students Zahra Tayebi, Sarwan Ali, Prakash Chourasia and Taslim Murad, and Master's students Babatunde Bello, Supriya Arun and Kaustubh Khandai.

- Babatunde Bello graduated with Master's thesis titled: "Functional enrichment analysis of transcriptomics data of breast cancer RNA-seq".

Fall 2021 PhD students Zahra Tayebi, Sarwan Ali and Prakash Chourasia, Master's students Babatunde Bello and Kaustubh Khandai, and Honors student Yijing Zhou.

- Sarwan Ali won an MBD fellowship, funding the remainder of his study.
- Yijing Zhou graduated with honors thesis titled: "Towards a complete workflow for inferring tumor phylogeny from raw single-cell sequencing reads"

Spring 2021 I supervised PhD students Zahra Tayebi and Sarwan Ali

Fall 2020 I supervised PhD student Zahra Tayebi

### Volunteer Experience

2017—2019 Research Advisor, Lycée Stendhal, Milan, Italy
Advising and guiding students on an after-school mathematics project which is part of the larger European program: Math en Jeans. There is an annual conference where the students go to present their work on the year's project.

## Grants

2024 Using AI techniques to uncover transmission dynamics and hotspots from HIV big data, Research Internal Grant (RIG), GSU, Amount: $20,000 USD (1 year)
PI: Alexander Kirpich

○ To collect HIV incidence data and numerically encode it for analysis for trends in transmission with machine learning algorithms

2024 REU site: Multidisciplinary graph data analytics, National Science Foundation, Amount: $372,720 USD (3 years)
PI: Esra Akbas, Mehmet Emin Aktas

○ To establish Georgia State University as a Research Experiences for Undergraduates (REU) site by hiring paid undergraduate summer interns to work together with faculty and graduate students on their research projects.

2021 Global education online module design, Hanyang University ERICA, Amount: $24,255 USD (3 years)
PI: Zhipeng Cai, Co-I: Armin Mikler, Rajshekhar Sunderraman, Pavel Skums, Murray Patterson, Rolando Estrada, Yingshu Li, Jennifer Sherer, Geoffrey Graybeal

○ For the development of online modules in data science and business intelligence, which could be offered as a course or training program.

### Contracts

2021 AI Learning Alliance: Broadening participation, Meta, Inc., Amount: $50,000 USD (2 years)
PI: Anu Bourgeois, Co-I: Murray Patterson

○ To support work related to providing education at the frontier of R&D and to increase the representation of historically marginalized researchers and engineers by preparing them for jobs in the artificial intelligence field.

### Under Review

2025 SCH: Tracing semantic shifts in textual and genomic data: A new approach for understanding the co-evolution of human behavior and pathogen phenotypic traits, National Science Foundation, Amount: $899,924 (3 years)
PI: Murray Patterson, Co-I: Alexander Zelikovsky, Mario Kubek, Alexander Kirpich and Gerardo Chowell

○ Human activity such as drug use and homelessness results in elevated susceptibility to infectious diseases, representing a critical global public health concern. Importantly, there is an ongoing co- evolution between human host behavior and pathogens: certain activities can accelerate pathogen transmission, thereby expanding opportunities for viral diversification and fostering the emergence of novel, potentially more transmissible or drug-resistant variants that threaten the broader population. Understanding and predicting such co-evolutionary events is essential for public health preparedness. We hypothesize that these events may manifest as synchronized semantic shifts in human-generated text (e.g., social media) and in the organizational or informational structure of viral genomes. Detecting these shifts independently is challenging, and their joint detection presents an even greater methodological problem. To address this, we propose a cross-domain language model (LM)-based framework designed to identify semantic shifts in both social discourse and epistatically-shaped viral genomes, and to infer associations between them.

2025 CAREER: Learning new patterns of evolution from sequencing big data, National Science Foundation, Amount: $651,363 USD (5 years)
PI: Murray Patterson

○ The current amount of molecular sequencing data available is orders of magnitude larger than in any other time throughout history, and has revealed evidence of evolutionary events such as saltations (evolutionary jumps) and reversions (back mutations). These patterns seem to contradict a century of development of evolutionary theory, suggesting a gap in our understanding of evolution, particularly at the smaller scale of viruses and tumors, where these events are often being observed. Better computational methods, models, and their resulting tools for the analysis of this big data resource are needed to close this gap. This would result in better surveillance, screening, and therapeutic strategies, since some viral strains of concern are the result of a saltation event from a more benign strain, while sequencing data from tumors reveal the widespread presence of reversions in any phylogenetic reconstruction from these sequences. This proposal develops methods based on phylo- and population genetics techniques and deep learning for better anticipating saltations and modeling reversions.

## Awards and Honors

2023 Nomination for membership to Sigma Xi, the scientific research honor society

2011 The Marie Curie Alain Bensoussan (ABCDE) Postdoctoral Fellowship of the European Research Consortium for Informatics and Mathematics (ERCIM)
Award of 60,300 € and 5000 € for travel, in order to fully fund 2 years of postdoctoral study in the European ERCIM member countries

2006 The Postgraduate Scholarship (PGS) of the Natural Sciences and Engineering Research Council (NSERC) of Canada
Award of $63,000 CAD in order to fully fund 3 years of postgraduate (PhD) research

2006 Simon Fraser University Computing Science Graduate Fellowship
Award of $3000 CAD towards funding postgraduate research

2005 Simon Fraser University Computing Science Graduate Fellowship
Award of $3000 CAD towards funding postgraduate research

2002 The Undergraduate Student Research Award (USRA) of the Natural Sciences and Engineering Research Council (NSERC) of Canada
Award of $6400 CAD in order to fully fund a summer of undergraduate research

## Technical Skills

| | | |
|---|---|---|
| advanced | C/C++, Python, LaTeX | 7+ years experience |
| intermediate | Java, Unix tools, Shell programming | 4+ years experience |
| basic | Matlab, R, Prolog, Haskell | 2+ years experience |

## Languages

| | | |
|---|---|---|
| English | maternal language | |
| French | second language | conversational and technical fluency |
| Italian | intermediate | comprehension and basic conversational skills |
| Spanish | intermediate | comprehension and basic conversational skills |
| 中文 | basic | basic words and phrases |

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## Publications

### Journal Articles

1. Taslim Murad, Prakash Chourasia, Sarwan Ali, Imdad Khan, and Murray Patterson. DANCE: Deep learning-assisted Analysis of proteiN sequences using Chaos Enhanced kaleidoscopic images. Journal of Computational Biology, 2026. to appear.
2. Ezekiel Ayotomiwa Adeniyi, Akshay Juyal, Pavel Skums, Murray Patterson, and Alex Zelikovsky. Uncovering epistatic interactions in SARS-CoV-2 evolution through hidden Markov models. Journal of Computational Biology, 2026. to appear.
3. Sarwan Ali, Prakash Chourasia, and Murray Patterson. Anderson acceleration for molecular sequencing. Neural Computing and Applications, 2025. to appear.
4. Sarwan Ali, Prakash Chourasia, Bipin Koirala, and Murray Patterson. Nearest neighbor CCP-based molecular sequence analysis. Transactions on Computational Biology and Bioinformatics, pages 1–14, 2025. doi:10.1109/TCBBIO.2025.3621138.
5. Simone Ciccolella, Murray Patterson, Iman Hajirasouliha, and Gianluca Della Vedova. Cancer progression inference using a finite-state model to allow recurrences and losses of mutations. Neural Computing and Applications, 2025. doi:10.1007/s00521-025-11474-1.
6. Sarwan Ali, Tamkanat E. Ali, and Murray Patterson. Hash-binding: DNA-protein binding prediction using hashing-based embedding. Neural Computing and Applications, 2025. doi:10.1007/s00521-025-11349-5.
7. Sarwan Ali, Tamkanat E. Ali, Haris Mansoor, Prakash Chourasia, and Murray Patterson. Hist2Vec: a histogram and kernel-based embedding method for molecular sequence analysis. Expert Systems with Applications, 273:126859, 2025. doi:10.1016/j.eswa.2025.126859.
8. Sarwan Ali, Madiha Shabbir, Haris Mansoor, Prakash Chourasia, and Murray Patterson. Elliptic geometry-based kernel matrix for improved biological sequence classification. Knowledge-Based Systems, 304:112479, 2024. doi:10.1016/j.knosys.2024.112479.
9. Zahra Tayebi, Sarwan Ali, and Murray Patterson. TCellR2Vec: Efficient feature selection for TCR sequences for cancer classification. PeerJ Computer Science, 10:e2239, 2024. doi:10.7717/peerj-cs.2239.
10. Sarwan Ali, Tamkanat E. Ali, Taslim Murad, Haris Mansoor, and Murray Patterson. Molecular sequence classification using efficient kernel based embedding. Information Sciences, 679:121100, 2024. doi:10.1016/j.ins.2024.121100.
11. Sarwan Ali, Prakash Chourasia, and Murray Patterson. From PDB files to protein features: a comparative analysis of PDB bind and STCRDAB datasets. Medical & Biological Engineering & Computing, 62:2449–2483, 2024. doi:10.1007/s11517-024-03074-3.

12. Zahra Tayebi, Sarwan Ali, Taslim Murad, Imdad Ullah Khan, and Murray Patterson. PseAAC2Vec protein encoding for TCR protein sequence classification. Computers in Biology and Medicine, 170:107956, 2024. doi:10.1016/j.compbiomed.2024.107956.
13. Sarwan Ali, Prakash Chourasia, and Murray Patterson. When protein structure embedding meets large language models. MDPI Genes, 15(1):25, 2023. doi:10.3390/genes15010025.
14. Sarwan Ali and Murray Patterson. Improving IOSOMAP efficiency with RKS: a comparative study with t-distributed neighbor embedding on protein sequences. MDPI J — Multidisciplinary Scientific Journal, 6(4):579–591, 2023. doi:10.3390/j6040038.
15. Taslim Murad*, Sarwan Ali*, Imdad Ullah Khan, and Murray Patterson. Spike2CGR: an efficient method for spike sequence classification using chaos game representation. Machine Learning, 112:3633–3658, 2023. doi:10.1007/s10994-023-06371-4.
16. Sarwan Ali, Prakash Chourasia, Zahra Tayebi, Babatunde Bello, and Murray Patterson. ViralVectors: compact and scalable alignment-free virome feature generation. Medical & Biological Engineering & Computing, 2023. doi:10.1007/s11517-023-02837-8.
17. Taslim Murad, Sarwan Ali, and Murray Patterson. Exploring the potential of GANs in biological sequence analysis. MDPI Biology, 12(6):854, 2023. doi:10.3390/biology12060854.
18. Bikram Sahoo*, Sarwan Ali, Pin-Yu Chen, Murray Patterson, and Alexander Zelikovsky*. Assessing the resilience of machine learning classification algorithms on SARS-CoV-2 genome sequences generated with long-read specific errors. Biomolecules, 13(6):934, 2023. doi:10.3390/biom13060934.
19. Danushka Bandara, Karen Exantus, Cristian Navarro-Martinez, Murray Patterson, and Ashley Byun. Identifying distinguishing acoustic features in felid vocalizations based on call type and species classification. Acoustics Australia, 2023. doi:10.1007/s40857-023-00298-5.
20. Sarwan Ali, Babatunde Bello, and Murray Patterson. Solvent accessibility of coronaviridae spike proteins through the lens of information gain. MDPI J — Multidisciplinary Scientific Journal, 6(2):236–247, 2023. doi:10.3390/j6020018.
21. Sarwan Ali, Bikram Sahoo, Alexander Zelikovsky, Pin-Yu Chen, and Murray Patterson. Benchmarking machine learning robustness in covid-19 genome sequence classification. Scientific Reports, 13(1):4154, 2023. doi:10.1038/s41598-023-31368-3.
22. Prakash Chourasia, Sarwan Ali, Simone Ciccolella, Gianluca Della Vedova, and Murray Patterson. Reads2Vec: Efficient embedding of raw high-throughput sequencing reads data. Journal of Computational Biology, 2022. doi:10.1089/cmb.2022.0424.
23. Sarwan Ali, Babatunde Bello, Zahra Tayebi, and Murray Patterson. Characterizing SARS-CoV-2 spike sequences based on geographical location. Journal of Computational Biology, 2022. doi:10.1089/cmb.2022.0391.

24. Sarwan Ali, Bikram Sahoo, Muhammad Asad Khan, Alexander Zelikovsky, Imdad Ullah Khan*, and Murray Patterson*. Efficient approximate kernel based spike sequence classification. Transactions on Computational Biology and Bioinformatics, 2022. doi:10.1109/TCBB.2022.3206284.

25. Kaustubh Khandai, Cristian Navarro-Martinez, Brendan Smith, Rebecca Buonopane, S. Ashley Byun*, and Murray Patterson*. Determining significant correlation between pairs of extant characters in a small parsimony framework. Journal of Computational Biology, 29(10):1132–1154, 2022. doi:10.1089/cmb.2022.0141.

26. Sarwan Ali, Babatunde Bello, Prakash Chourasia, Ria Thazhe Punathil, Yijing Zhou, and Murray Patterson. PWM2Vec: An efficient embedding approach for viral host specification from coronavirus spike sequences. MDPI Biology, 11(3):418, 2022. doi:10.3390/biology11030418.

27. Sarwan Ali, Yijing Zhou, and Murray Patterson. Efficient analysis of COVID-19 clinical data using machine learning models. Medical & Biological Engineering & Computing, 60(7):1881–1896, 2021. doi:10.1007/s11517-022-02570-8.

28. Zahra Tayebi, Sarwan Ali, and Murray Patterson. Robust representation and efficient feature selection allows for effective clustering of SARS-CoV-2 variants. Algorithms, 14(12):348, 2021. doi:10.3390/a14120348.

29. Sarwan Ali, Simone Ciccolella, Lorenzo Lucarella, Gianluca Della Vedova, and Murray Patterson. Simpler and faster development of tumor phylogeny pipelines. Journal of Computational Biology, 28(11):1142–1155, 2021. doi:10.1089/cmb.2021.0271.

30. Andrew Melnyk, Fatemeh Mohebbi, Sergey Knyazev, Bikram Sahoo, Roya Hosseini, Pavel Skums, Alex Zelikovsky*, and Murray Patterson*. From alpha to zeta: Identifying variants and subtypes of SARS-CoV-2 via clustering. Journal of Computational Biology, 28(11):1113–1129, 2021. doi:10.1089/cmb.2021.0302.

31. Simone Ciccolella*, Murray Patterson*, Paola Bonizzoni, and Gianluca Della Vedova. Effective clustering for single cell sequencing cancer data. IEEE Journal of Biomedical and Health Informatics, 25(11):4068–4078, 2021. doi:10.1109/JBHI.2021.3081380.

32. Simone Ciccolella, Mauricio Soto, Murray D. Patterson, Gianluca Della Vedova, Iman Hajirasouliha, and Paola Bonizzoni. Gpps: An ILP-based approach for inferring cancer progression with mutation losses from single cell data. BMC Bioinformatics, 21(1):413, 2020. doi:10.1186/s12859-020-03736-7.

33. Simone Ciccolella, Camir Ricketts, Mauricio Soto Gomez, Murray Patterson, Dana Silverbush, Paola Bonizzoni, Iman Hajirasouliha, and Gianluca Della Vedova. Inferring cancer progression from single-cell sequencing while allowing mutation losses. Bioinformatics, 37(3):326–333, 2020. doi:10.1093/bioinformatics/btaa722.

34. Raffaella Rizzi, Stefano Beretta, Murray Patterson, Yuri Pirola, Marco Previtali, Gianluca Della Vedova, and Paola Bonizzoni. Overlap graphs and de Bruijn graphs: data structures for de novo genome assembly in the big data era. Quantitative Biology, 7(4):278–292, 2019. doi:10.1007/s40484-019-0181-x.

35. Stefano Beretta*, Murray Patterson*, Simone Zaccaria, Gianluca Della Vedova, and Paola Bonizzoni. HapCHAT: Adaptive haplotype assembly for efficiently leveraging high coverage in long reads. BMC Bioinformatics, 19(1):252, 2018. doi:10.1186/s12859-018-2253-8.

36. Wandrille Duchemin, Yoann Anselmetti, Murray Patterson, Yann Ponty, Sèverine Bérard, Cedric Chauve, Celine Scornavacca, Vincent Daubin, and Eric Tannier. DeCoSTAR: Reconstructing the ancestral organization of genes or genomes using reconciled phylogenies. Genome Biology and Evolution, 9(5):1312–1319, 2017. doi:10.1093/gbe/evx069.

37. Andrea Bracciali, Marco Aldinucci, Murray Patterson, Tobias Marschall, Nadia Pisanti, Ivan Merelli, and Massimo Torquati. PWhatsHap: Efficient haplotyping for future generation sequencing. BMC Bioinformatics, 17(11):342, 2016. doi:10.1186/s12859-016-1170-y.

38. Murray Patterson*, Tobias Marschall*, Nadia Pisanti, Leo van Iersel, Leen Stougie, Gunnar W. Klau†, and Alexander Schoenhuth†. WhatsHap: Weighted haplotype assembly for future-generation sequencing reads. Journal of Computational Biology, 22(6):498–509, 2015. doi:10.1089/cmb.2014.0157.

39. Murray Patterson, Gergely Szöllősi, Vincent Daubin, and Eric Tannier. Lateral gene transfer, rearrangement, reconciliation. BMC Bioinformatics, 14(15):S4, 2013. doi:10.1186/1471-2105-14-S15-S4.

40. Mohammed El-Kebir*, Tobias Marschall*, Inken Wohlers*, Murray Patterson, Jaap Heringa, Alexander Schoenhuth, and Gunnar W. Klau. Mapping proteins in the presence of paralogs using units of coevolution. BMC Bioinformatics, 14(15):S18, 2013. doi:10.1186/1471-2105-14-S15-S18.

41. Ján Maňuch*, Murray Patterson*, Roland Wittler*, Cedric Chauve, and Eric Tannier. Linearization of ancestral multichromosomal genomes. BMC Bioinformatics, 13(19):S11, 2012. doi:10.1186/1471-2105-13-S19-S11.

42. Ján Maňuch*, Murray Patterson*, and Cedric Chauve. Hardness results on the gapped consecutive-ones property problem. Discrete Applied Mathematics, 160(18):2760–2768, 2012. doi:10.1016/j.dam.2012.03.019.

43. Ján Maňuch* and Murray Patterson*. The complexity of the gapped consecutive-ones property problem for matrices of bounded maximum degree. Journal of Computational Biology, 18(9):1243–1253, 2011. doi:10.1089/cmb.2011.0128.

44. Roland Wittler, Ján Maňuch*, Murray Patterson*, and Jens Stoye. Consistency of sequence-based gene clusters. Journal of Computational Biology, 18(9):1023–1039, 2011. doi:10.1089/cmb.2011.0083.

## Conference Proceedings

45. Chandra Kiran Guntupalli, Thomas Böhme, Murray Patterson, and Mario M. Kubek. Meaning in motion: tracking temporal semantic drift through incremental representation spaces. In the 8th International Conference on Natural Language Processing (ICNLP, Xi'an, China, 2026). 2026. to appear.
46. Ahmed Luqman, Khuzemah Qazi, Murray Patterson, Malik Jahan Khan, and Imdadullah Khan. A data-free analytical quantization scheme for deep learning models. In the 2025 IEEE International Conference on Data Mining (IEEE ICDM, Washington DC, USA, 2025). 2025. to appear.
47. Sarwan Ali, Taslim Murad, Prakash Chourasia, Haris Mansoor, Imdad Ullah Khan, Pin-Yu Chen, and Murray Patterson. Position specific scoring is all you need? Revisiting protein sequence classification tasks. In the 22nd RECOMB Conference on Comparative Genomics (RECOMB-CG, Seoul, South Korea, 2025), 202–215. 2026. doi:10.1007/978-3-031-94928-9_15.
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